

Automated All-Weather Foreign Object Debris (FOD) Detection System for Runway Safety – “HAWKEYE”

Need for FOD Detection?

FODs can get ingested into engines or get lodged into mechanisms damaging aircraft leading to **Catastrophic Loss of Life****Increased Maintenance & Operational Costs**

Issue with Conventional Detection Methods

- ❑ **Manual Removal:** Human confounding factors – *visibility, accuracy, efficiency focus, motivation* every individual will be different
- ❑ **MRS (Mechanical Runway Sweeper):** (i) *Slow Speed* (5-10km/hour), (ii) Area of coverage of FOD retrieval is minimal confining to the *area of the truck*, (iii) To cover the breadth of 150ft of the runway *multiple trips to cover the area*, (iv) If a *FOD is dislodged from the MRS itself* there is no retrieval mechanism

AIM:

Develop an **Automated & Cost-Effective FOD Detection System** capable of functioning under **all Weather & all Visibility conditions**



FOD sensors – Front

FOD sensors – Back

FOD Sensors	Mobile Hybrid → Radar + Vision
Radar	Highly Directional Millimeter-Wave Radio Transceiver (76-81GHz)
Vision	High-Resolution RGB & NIR Camera with pan & zoom
Vehicle Speed	Max 50km/hr
Time for 3km Runway	< 4 minutes

- ✓ **Multiple-object detection**
- ✓ **No Blind-spots**, runway scanned in full redundancy
- ✓ **Back sensors** for additional accuracy & vehicle debris and cancelling reflection errors

State-of-the-Art Data Processing Methodologies

- ❑ **Fast-Object Detection Algorithms**
 - YOLO V5, R-CNN, DETR – Tracking and Pipelining ODAs
 - CSRT is the base algorithm used
- ❑ **Deep Learning based intelligent system**
 - Life-long learning mechanism enabling predictions to optimize response time

Personnel Required

Alarm/Alert
in case one or more FOD is detected

Manual FOD Retrieval

2 → Driver + Pickup Operator

Pickup Operator can look at the **camera footage** on the console inside the car and use zoom/pan options to **locate the FOD** on the runway

Once the FOD is located, pickup operator can climb down from the vehicle, collect the FOD and store as per surveillance regulations



MOBILE OPERATOR CONSOLE

VEHICLE MOUNTING OF CAMERA AND DATA COLLECTION



RESULTS OF FOD DETECTION ON RUNWAY USING DIFFERENT CAMERAS & ARTIFICIAL INTELLIGENCE TRAINED COMPUTER MODEL

Disadvantages of thermal cameras

1. Low frame rate – Misses small FOD.
2. Small FOV
3. Higher resolution often lowers frame rate.
4. Data processing load- AI Model, Data incompatible
5. False negatives/positives from surface temperature variations, shadows, or tire marks
6. Over heating of cameras processor
7. Export restrictions apply on >60 Hz IR cameras.
8. Extremely Costly per unit



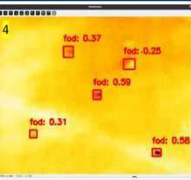
Day- FOD Detection using DSLR



Night - FOD Detection using DSLR



FOD Detection using HHTI Camera



FOD Detection using LWIR Camera



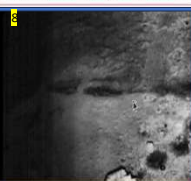
FOD Detection using SWIR camera



FOD Detection using MWIR camera



FOD Detection using NWIR camera

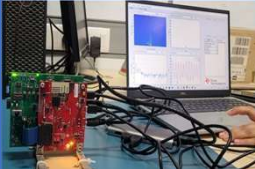


NIR camera with UV Filters

FINAL PROTOTYPE DEVELOPED



RESULTS OF MM WAVE RADAR FOD DETECTION





Object	Dimensions (cm)	SNR(dB)	Range (m)
Coin size plate	D=2.5	40.13, 40.09	0.4

Mm wave Radar Detection

1. Mm Wave Radar 77 GHz in Lab
2. Vehicle mounted with Mm Wave radar from window
3. FOD Identified using Mm wave radar signal